

AMBERIEU-EN-BUGEY AB, France

WMO: 074820

Lat: 45.9765N

Lon: 5.3293E

Elev: 823

StdP: 14.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	22.4	14.1	11.6	20.7	18.4	14.3	25.2	23.2	41.6	20.8	43.1	4.5	10	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.0	91.2	69.1	87.5	68.3	84.2	67.4	71.2	86.0	69.8	83.9	68.4	81.5	8.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	100.4	77.1	65.1	95.6	75.2	63.7	91.2	73.5	35.4	86.1	34.2	84.0	33.1	81.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.6	17.8	15.5	DB	12.8	95.5	5.6	3.9	8.8	98.3	5.5	100.6	2.4	102.8	-1.7	105.6	(4)
(5)			WB	12.4	74.0	5.5	1.4	8.5	75.0	5.2	75.8	2.1	76.6	-1.9	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	37.6	39.5	46.2	52.2	59.3	66.2	69.8	68.9	61.5	54.7	44.6	38.7		
	DBStd	13.46	7.95	7.84	6.93	6.55	6.46	6.80	6.26	6.38	6.22	7.26	7.98	8.12		
	HDD50	1497	389	300	157	52	5	0	0	0	1	37	198	358		
	HDD65	4744	850	715	583	386	197	66	24	30	141	323	613	816		
	CDD50	2718	3	5	39	117	295	487	613	587	345	184	36	7		
	CDD65	489	0	0	0	1	22	103	172	152	34	5	0	0		
	CDH74	5130	0	0	0	35	240	1101	1830	1585	320	19	0	0		
	CDH80	1823	0	0	0	2	39	365	700	652	65	0	0	0		
(14) Wind	WSAvg	6.3	6.2	6.8	7.2	7.3	6.6	6.5	6.0	5.7	5.6	5.8	6.2	6.1		
(15) Precipitation	PrecAvg	45.4	3.5	3.2	2.9	3.7	4.1	3.5	3.4	3.3	4.4	4.8	4.9	3.8		
	PrecMax	55.6	6.9	7.2	10.0	8.2	8.6	6.8	9.0	8.1	13.0	10.0	12.9	7.3		
	PrecMin	32.0	1.1	0.5	0.5	0.7	0.8	0.9	1.1	0.9	1.1	1.0	2.0	0.6		
	PrecStd	6.1	1.6	1.9	2.0	2.0	1.9	1.5	1.8	1.6	2.9	2.0	2.6	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	61.8	70.5	78.7	84.3	93.0	95.1	97.6	86.2	76.4	66.5	59.3		
		MCWB	50.8	49.7	54.6	60.0	65.7	70.3	69.2	69.7	66.4	63.0	56.5	51.6		
	2%	DB	54.1	57.5	66.0	73.0	80.1	87.8	91.0	91.4	81.5	72.8	62.7	56.0		
		MCWB	48.6	48.3	52.8	57.7	62.8	68.5	69.1	69.1	66.1	61.9	54.5	50.1		
	5%	DB	51.9	54.3	62.3	69.2	76.3	84.1	87.3	87.1	77.7	69.4	59.8	53.7		
		MCWB	47.3	47.0	51.3	55.8	61.7	67.5	68.5	68.2	64.2	60.3	53.4	48.8		
	10%	DB	49.3	51.4	58.6	65.4	72.7	80.6	83.7	82.8	74.0	66.3	56.7	51.0		
		MCWB	45.4	45.7	49.7	54.0	60.2	65.6	67.3	67.3	62.9	58.7	51.9	47.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	51.3	56.0	61.7	67.4	73.2	73.0	72.7	70.0	64.7	58.0	53.0		
		MCDB	55.8	57.8	67.1	75.9	79.9	87.2	87.6	89.2	80.0	72.9	64.3	56.9		
	2%	WB	49.5	49.4	53.9	58.8	65.1	70.5	71.1	70.8	67.1	62.9	55.6	50.8		
		MCDB	53.3	55.5	62.8	70.8	77.0	84.2	86.1	86.4	78.0	70.1	61.0	54.9		
	5%	WB	47.6	47.8	52.2	56.6	62.8	68.5	69.7	69.3	65.4	61.3	53.9	49.2		
		MCDB	51.4	53.4	60.4	67.3	73.2	81.5	83.9	83.5	75.5	67.9	58.8	53.0		
	10%	WB	45.6	46.0	50.4	54.7	60.8	66.7	68.3	67.9	63.6	59.4	52.2	47.3		
		MCDB	48.9	50.9	57.8	63.8	70.5	78.4	81.6	80.7	72.1	65.4	56.4	50.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	14.6	19.1	20.1	19.6	21.2	22.0	21.9	19.9	16.4	12.8	10.8		
		MCDBR	13.7	20.6	25.0	27.1	25.3	25.9	27.1	28.9	25.6	20.5	17.0	14.1		
		MCWBR	9.3	13.0	14.0	14.0	12.3	11.2	10.6	10.8	12.0	11.1	10.6	9.6		
	5% WB	MCDBR	12.8	18.3	22.3	24.8	22.9	24.7	24.5	25.4	22.4	18.4	15.5	12.5		
		MCWBR	9.1	11.9	12.9	13.2	12.0	11.2	10.5	10.6	10.9	10.4	10.1	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.329	0.360	0.396	0.403	0.404	0.401	0.394	0.377	0.362	0.329	0.306		
	taud		2.463	2.406	2.350	2.259	2.289	2.320	2.326	2.354	2.385	2.423	2.469	2.472		
	Ebn at Noon		253	268	274	273	275	274	273	271	266	252	242	240		
	Edh at Noon		22	28	34	40	41	40	39	36	32	27	22	19		
(44) All-Sky Solar Radiation	RadAvg		385	653	1065	1430	1669	1983	1930	1669	1278	796	436	320		
	RadStd		46	95	134	189	163	164	165	128	99	75	46	52		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page